

MYMENSINGH GIRLS' CADET COLLEGE
PROGRESS TEST EXAMINATION - 2026
CLASS: HSC
STATISTICS (CREATIVE)
SECOND PAPER
[According to the Syllabus of 2026]
TIME – 1 hours & 25 minutes
FULL MARKS – 30

Subject Code:

1	3	0
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[N.B. – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **THREE** questions taking at least ONE question from each group]

Group - A

1. In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected.
- (a) What does $P(A) = 0$ mean?

1
- (b) Find the probability that the student passed in at least one subject.

2
- (c) What is the probability that the student passed in both the subjects?

3
- (d) Determine the probability that the student passed in only one subject.

4
2. The monthly electricity bills (in dollars) of 5 households are 45, 60, 55, 70, and 50. A researcher claims that if random samples of size 3 are drawn from these bills without replacement, the sample mean will provide an unbiased estimate of the population mean.
- (a) Define a sample in this context.

1
- (b) State two drawbacks of conducting a census.

2
- (c) Compute the population variance from the given data.

3
- (d) Verify the researcher’s claim regarding unbiasedness of the sample mean.

4

Group - B

3. The joint probability function of two random variables X and Y is given by:

$$P(X, Y) = \frac{x + y + 1}{42}; \quad x = 0, 1, 2; \quad y = 0, 1, 2, 3$$

- (a) What is a discrete random variable?

1
- (b) What are the required properties of a probability distribution?

2
- (c) Calculate the marginal probability $P(Y)$.

3
- (d) Determine $P(Y|X = 1)$ and $P(Y|X = 0)$.

4
4. The probability distribution of a random X is provided below:

X	-1	0	1	2	3
P(x)	$\frac{3}{20}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{20}$

- (a) What is the maximum possible value of variance?

1
- (b) Find $E(X)$ from the stem.

2
- (c) Find $E(W)$, where $W = X^2$.

3
- (d) Find Variance of $(\frac{2}{3}X + 3)$.

4

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Group - A

1. **A and B are two events where $P(A) = \frac{1}{3}, P(B) = \frac{3}{4}$, and $P(A \cup B) = \frac{10}{12}$. A professor said that $P(A|B) = P(B|A)$.**
- (a) What is the minimum value of probability? 1
 - (b) Why can't probability be negative? Explain with an example or theoretically. 2
 - (c) Find $P(A \cap B)$ from the stem. 3
 - (d) Evaluate the professor's assertion. 4

2. **The joint probability function of two random variables X and Y is described by:**

$$P(X, Y) = \frac{2x + 3y}{45}; \quad x = 0, 1, 2; \quad y = 0, 1, 2$$

- (a) Write down the formula for conditional probability. 1
 - (b) What is the relationship between marginal and joint probability? 2
 - (c) Find $P(X)$. 3
 - (d) Find $P(X|Y)$ and $P(X|Y = 0)$. 4
3. **A professor showed a probability distribution in a class. The distribution is given below. He informed that the value of the arithmetic mean of the distribution is 3.**

x	1	2	3	4	5
p(x)	0.1	a	0.3	b	0.2

- (a) What is the formula of expectation? 1
 - (b) What is the variance of a constant? Explain logically. 2
 - (c) What are the values of a & b? 3
 - (d) Find and explain the variance of the distribution. 4
4. **The scores of 4 students in an IELTS Test are 7.5, 7.0, 8.5, and 6.0. An instructor said, if samples of size 2 are selected from these score without replacement, the sample mean will be an unbiased estimate of the population mean.**
- (a) What is a sample? 1
 - (b) What are the disadvantages of a census? 2
 - (c) Estimate the population variance from the data. 3
 - (d) Evaluate the statement of the instructor. 4

Group - B

5. **In a binomial distribution, the mean and standard deviations are 3 and $\sqrt{\frac{3}{2}}$. An analysis revealed that the distribution is symmetric.**
- (a) How many outcomes are there in a Bernoulli trial? 1
 - (b) Explain the relationship between the mean and variance of the binomial distribution. 2

- (c) Find the value of $P(X = 2)$. 3
 - (d) Assess the finding of the analysis. 4
6. **The number of device failures within a certain period in a tech industry follows a Poisson distribution with a standard deviation of 2.5.**
- (a) What is e in Poisson distribution? 1
 - (b) In a Poisson distribution, $P(2) = P(3)$. What is its standard deviation? 2
 - (c) Find $P(X \geq 3)$. 3
 - (d) What is the probabiility that there are no more than 2 device failures within the given period? 4

7. **Price and Quantity Data for Rice, Pulse, and Oil in the years 2013 and 2015 are given below.**

Commodity	(2010)		(2011)	
	Price	Quantity	Price	Quantity
A	45	30	50	40
B	105	5	110	8
C	85	2	80	5

Table 1: Price and Quantity Data for Three Commodities

- (a) What is an ideal index number? 1
 - (b) Is index number unit-free? 2
 - (c) Compute the price index number, considering 2013 as the base year, using simple aggregate method. 3
 - (d) Sift the time reversal test for the Laspeyres’ price index number. 4
8. **The following dataset records the number of women in different age groups and their respective live births as part of a demographic study.**

Age Group	15–19	20–24	25–29	30–34	35–39	40–44	45–49
No. of Women	540	760	530	495	450	505	430
No. of Live Births	109	198	86	90	65	76	60

- (a) Write down the formula of GRR. 1
- (b) Differentiate between GRR and NRR. 2
- (c) Determine the Age Specific Birth Rates (ASFR) for the given age groups. 3
- (d) Find the General Fertility Rate (GFR) and explain its significance in relation to ASFR. 4